

EDUCATION

Université de Toulouse, Toulouse, France

M.S. Interactions of Computer Science and Mathematics for AI 2025–2026

National Institute of Technology Silchar, Silchar, India

B.Tech. Electronics and Communication Engineering 2021–2025

RESEARCH EXPERIENCE

Institut de Recherche en Informatique de Toulouse (IRIT), France

Research Intern (Full-time) May 2024–July 2024, Jan 2025–Mar 2025

- Researched discourse reasoning and numerical comprehension in large language models under Dr. Nicholas Asher
- Fine-tuned and evaluated GPT-style architectures (125M params) on multi-GPU HPC clusters for text + numeric joint reasoning
- Implemented reproducible experiments using PyTorch, DDP, and automated evaluation pipelines

Artificial Intelligence Institute, University of South Carolina

Research Intern (Remote) Jan 2023–Dec 2024

- Conducted large-scale evaluation across 15 LLMs to analyze reasoning failures and generative bias patterns
- Designed pipelines for statistical testing and task-specific evaluation
- Conducted large-scale experiments on visual paraphrasing and watermarking robustness, highlighting vulnerabilities of current methods
- Published award-winning findings at EMNLP 2023 (Outstanding Paper Award)

PUBLICATIONS

Counter Turing Test CT^2 : AI-Generated Text Detection is Not as Easy as You May Think – Introducing AI Detectability Index EMNLP 2023

M. Chakraborty, S. M. T. I. Tonmoy, S. M. M. Zaman, K. Sharma, N. R. Barman, C. Gupta, S. Gautam, T. Kumar, V. Jain, A. Chadha, A. P. Sheth, and A. Das

Outstanding Paper Award

DIMSUM: Discourse in Mathematical Reasoning as a Supervision Module SIGDIAL 2025

N. R. Barman, K. Sharma, N. Asher, and A. Chaturvedi

The Brittleness of Image Watermarking Techniques: Investigating Visual Paraphrasing for De-Watermarking AI-Generated Images arXiv 2024

N. R. Barman, K. Sharma, A. Aziz, S. Bajpai, S. Biswas, A. Chadha, V. Sharma, and A. Das

A Transformer-Based Approach to Automate Disease Prediction from Patient Descriptions IEEE CICT 2023

N. R. Barman, K. Sharma, and R. Hazra

Addressing Hate Speech: ATLANTIS for Efficient Hate Span Detection FIRE 2023

N. R. Barman, K. Sharma, Y. Poddar, A. Vetagiri, and P. Pakray

GRANTS AWARDED	Université de Toulouse ANITI Graduate School Master Excellence Scholarship	2025–2026
TECHNICAL SKILLS	Languages: Python, C++, Bash Frameworks: PyTorch, Transformers, TensorFlow, JAX, Scikit-learn, FastAPI Tools: Git, Docker, Linux Shell, Slurm, CUDA Platforms: Linux, AWS Miscellaneous: L ^A T _E X	
OPEN SOURCE CONTRIBUTIONS	pytorch/examples: Implemented Vision Transformer from scratch using only native PyTorch libraries, trained and evaluated on CIFAR-10 dataset huggingface/optimum: Removed TF models export support from optimum-onnx as it is not supported in Optimum v2	
SELECTED PROJECTS	automanim <i>FastAPI, Next.js, Manim, LLMs</i> Developed a conversational interface for generating and rendering Manim animations using large language models. Implemented live preview functionality, multi-turn conversation memory, and per-scene video configuration. Integrated multiple LLM backends including Ollama and Hugging Face Transformers for flexible deployment.	Nov 2025
POSITIONS OF RESPONSIBILITY	EMNLP Presenter <i>Conference Presentation</i> Presented Counter Turing Test (CT²) — Outstanding Paper Award Winner (Resources and Evaluation Track) at EMNLP 2023 Main Poster Session Teaching Assistant/Mentor <i>ML Club, NIT Silchar</i> Conducted Python and Machine Learning sessions for freshmen and sophomores, facilitating peer learning and mentorship	Dec 2023 Jun 2023–May 2025